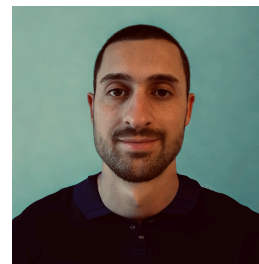


Igor Martinelli

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EDUCATION

ETH ZURICH | MSc IN COMPUTER SCIENCE

Sep. 2023 – Mar. 2026

Major: Machine Intelligence | Minor: Data Management Systems

ETH ZURICH | BSc IN COMPUTER SCIENCE

Sep. 2019 – Dec. 2023

EXPERIENCE

ELCA INFORMATIQUE | MACHINE LEARNING INTERN

Apr. 2026 – Present | Zurich, Switzerland

- Developed an end-to-end analytics and machine learning platform for electric bus fleets.
- Built explainable AI solutions, delivering predictive energy analytics, real-time fleet monitoring, and AI-generated operational insights through interactive dashboards.

BLP DIGITAL | DATA ANNOTATION INTERN

Apr. 2023 – Sep. 2023 | Zurich, Switzerland

- Annotated and curated high-quality datasets for training complex computer vision and deep learning models and trained new interns.

PROJECTS

MASTER THESIS | ENHANCING PROSTHETIC AND ROBOTIC DEVICES WITH NEUROMORPHIC SPIKING NEURAL NETWORKS FOR TACTILE OBJECT CLASSIFICATION

Sep. 2025 – Mar. 2026 | ETH Zurich

- Built a hardware-aware SNN pipeline encoding tactile pressure into sparse event streams for low-power robotics.
- Applied input compression and synaptic sparsity to meet chip routing constraints, reaching 86.97% accuracy on 17 classes.

PUBLICATIONS

IEEE BIOCAS | ACCEPTED 2026

Martinelli, I., Wang, X. & Donati, E. "Event-Driven Tactile Object Classification with Spiking Neural Networks and Input Compression."

NPJ BIOSENSING | JUNE 2024

Elgendi, M., Martinelli, I. & Menon, C. "Optimal signal quality index for remote photoplethysmogram sensing."

<https://doi.org/10.1038/s44328-024-00002-1>

EXTRACURRICULAR ACTIVITIES

CBR LOCARNO | LIFEGUARD

Summer 2016 – 2022 | Locarno, Switzerland

US ASCONA | GYMNASTICS COACH

Sept 2016 – Aug 2019 | Ascona, Switzerland

SKILLS

LANGUAGES

Python • C/C++ • SQL
Java • TypeScript • JavaScript
R • Swift

ML & AI

PyTorch • TensorFlow • CUDA
snntorch • Scikit-learn • SHAP
Azure ML • MLflow

DEVELOPER TOOLS

Docker • Git • VSCode
Microsoft Foundry • Streamlit

OPERATING SYSTEMS

Linux • UNIX • Windows

LANGUAGES

Italian: Native
English: Fluent
German: Professional
French: Basic

COURSEWORK

Deep Learning
Probabilistic Artificial Intelligence
Machine Perception
Natural Language Processing
Data Management Systems
Game Programming Lab

HOBBIES

Long distance running
Weightlifting

LINKS

Github:// [tinee29](#)
LinkedIn:// [igor-martinelli](#)